AI-Based Automated Evaluation of Handwritten Descriptive Answer Scripts Using Vision-Language OCR and Constraint-Aware LLM Grading

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***Abstract*—Evaluation of handwritten descriptive answer scripts has always been considered difficult due to handwriting differences, OCR errors, semantics variability of students' answers, and specific rules for grading in the particular examination process. The existing approaches to automated assessments mainly involve using the clean digital version of text and lexical matching techniques, restricting their application in real-life scenarios of handwritten examinations. This paper proposes an end-to-end AI-powered system that combines Vision-Language Optical Character Recognition (OCR) and constraint-aware Large Language Model (LLM) in order to provide semantic grading of handwritten answer scripts. A Florence-2 vision-language model that was fine-tuned on the IAM Handwriting Database is used for recognizing handwritten texts in a reliable way followed by an answer alignment algorithm that aligns the answers to questions considering the peculiarities of the examination process structure. The grading is performed by locally running Llama 3 (8B) grading engine that evaluates the answers according to the grading rubrics, including the marking scheme, optional questions, and section-wise grading.**

***Index Terms*—Automated Essay Scoring (AES), Handwritten Text Recognition, Vision-Language Models, Florence-2, Large Language Models, Semantic Answer Assessment, Constraint-Aware Grading, Educational AI.**

I. Introduction

Descriptive tests through handwriting still form one of the main methods of evaluating the students' conceptual knowledge, analytical skills, and problem-solving skills in education centers. Nonetheless, manual assessment of such handwritten answer papers involves intensive work, delays, and is always subjective, leading to inconsistent scoring and late posting of the results. As the number of students keeps increasing, there is an increased need for intelligent automated systems for conducting assessments [1], [14], [16].

The latest developments in NLP, VLMs, and LLMs have greatly enhanced Automated Essay Scoring and document understanding [5], [10], [18]. However, most AES tools are designed for digitally typed texts or make use of lexical matching and thus do not apply to handwritten descriptive exams where variations in handwriting styles, OCR mistakes, and different ways of writing an answer continue to be a problem [1], [9], [13]. In addition, existing methods hardly combine handwritten text recognition with semantic grading and take into account examination-specific marking criteria.

In order to overcome such shortcomings, this paper presents an end-to-end AI-based framework that integrates a Florence-2 Vision-Language OCR model with a constraint-aware Llama 3 (8B) grading engine for automated evaluation of handwritten descriptive answers. In this way, Florence-2 will be trained on the IAM Handwriting Database in order to increase the performance of OCR of handwritten text regardless of the diversity in writing style and conditions of documents. Automatically aligned responses will be semantically evaluated against predefined answers set by instructors. Grading will also take into account examination-related constraints such as marking schemes and section-wise allocation of marks.

The major contributions of this work are summarized as follows:

- An end-to-end framework integrating Florence-2 Vision-Language OCR with a constraint-aware Llama 3 grading engine for handwritten descriptive answer evaluation.
- An automated answer-alignment mechanism capable of handling complex examination layouts and optional-question structures.
- A rubric-guided semantic grading strategy that combines contextual reasoning with examination-specific grading constraints.
- Experimental evaluation on the IAM Handwriting Database and a curated handwritten examination dataset,

demonstrating the effectiveness of the proposed framework for practical educational assessment.

II. Literature Survey

The automated evaluation of handwritten descriptive answer scripts is an important research topic that lies at the intersection of computer vision, natural language processing, and educational technology. In earlier Automated Essay Scoring (AES) systems, the use of lexical features and manually crafted rules was predominant which prevented them from assessing semantic comprehension. Recent progress in Vision-Language Models (VLMs), handwriting recognition, and Large Language Models (LLMs) has made it possible to improve document understanding and semantic evaluation significantly. However, current techniques do not integrate the tasks of handwriting recognition and answer evaluation, taking into account examination specific constraints and rubric based evaluation. This work tries to overcome this problem by providing an integrated system for the end-to-end evaluation of handwritten descriptive answer scripts.

A. Automated Essay and Answer-Sheet Scoring with NLP and LLMs

First-generation AES systems made use of lexical properties, rules designed by humans, and keyword matching for mimicking human evaluation [1], [14]. While efficient from the computational point of view, these methods did not have the capacity to incorporate semantic comprehension and conceptual accuracy [13]. Syntactic and linguistic features used in machine learning algorithms enhanced consistency of scoring yet were task-specific and lacked generalizability [6], [15], [18].

The introduction of Transformer-based language models has greatly contributed to the automation of assessment via contextual semantic reasoning. Transformer-based AES models have been shown to exhibit high agreement with human assessors [4], [10], [12], and the recent use of LLaMA-based models has facilitated rubric-driven assessment via semantic analysis and explanation [2], [7], [8].

Nevertheless, existing grading systems based on LLMs make use of digitized data [5], [16] and disregard problems associated with handwriting. Errors in OCR adversely affect subsequent analysis on the semantic level, thereby decreasing the accuracy of grading. In spite of the fact that some recent works have considered multi-modal handwritten answer sheet assessment [20], Vision-Language OCR with constraint-aware LLM grading has not been sufficiently studied yet.

B. Vision-Language Models for Handwritten Document Understanding

Automated comprehension of handwritten documents still continues to be a key problem in automated educational assessment because of differences in handwriting, document quality, and page layout complexity. Traditional HTR systems yielded encouraging results on standard datasets such as the IAM Handwriting Database [13], [18], although their effectiveness dropped significantly in the case of handwritten examination documents with noise text and drawings [16].

TABLE I
Comparison of Existing Methods and the Proposed Framework

Feature	Existing	Proposed
Semantic Grading	Partial	✓
Handwritten OCR	Partial	✓
Constraint Handling	×	✓
Offline Deployment	×	✓
Queue Processing	×	✓
Human-in-the-loop OCR	×	✓

The emergence of Vision-Language Models (VLMs) has led to an increased efficiency in document understanding owing to the combined modeling of visual and textual data. In contrast to the traditional OCR pipeline that separates these two types of data, vision-language models such as Florence-2 are capable of performing end-to-end text extraction through prompting and are more robust to variations in handwriting and document layout [20].

Even with all these advancements, current literature mainly concentrates on handwriting recognition technology instead of the use of these technologies in AI-based marking solutions. The presented approach seeks to fill this gap by merging the Vision-Language OCR system with the constraint-aware LLM grading approach.

C. Research Gap

Current automated grading systems concentrate on either one or the other of these two aspects, but hardly ever have an integrated approach. Furthermore, constraint-based grading, offline operation, and scalability in handwriting evaluation of descriptive answers are not commonly supported [13], [16], [18].

III. Proposed Framework

In the proposed framework, an end-to-end framework of AI-driven approach to automate the assessment of handwritten descriptive answer scripts is proposed based on the combination of Vision-Language OCR, semantic reasoning using Large Language Model (LLM), and asynchronous inference management. Contrary to the traditional method where the task of handwritten text recognition and answer assessment is performed independently, the proposed framework creates an end-to-end processing pipeline to convert handwritten examination documents to structured texts for semantic assessment.

Evaluation is a multi-stage process that starts from scanning of question papers and handwritten answer sheets in PDF or image format. Handwritten document understanding using the Florence-2 Vision-Language OCR component involves extraction of the textual content while keeping the context and structure of the original document intact. Subsequently, the extracted text goes through the Llama 3-based semantic evaluation engine that evaluates the conceptual similarity of the answers provided by the students to the answers given

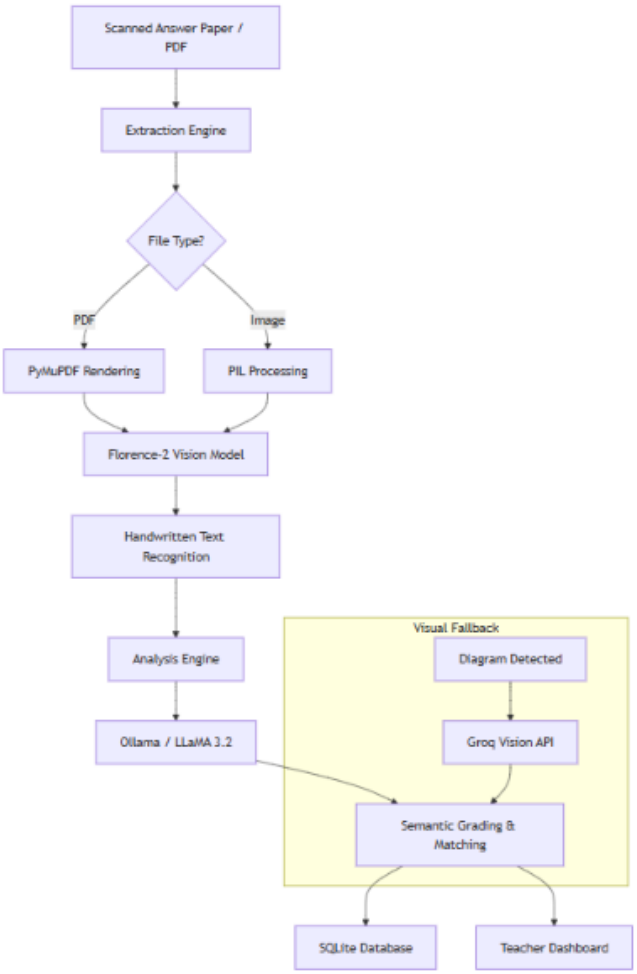


Fig. 1. Overall architecture of the proposed evaluation framework.

by the instructor. To facilitate the batch processing of a high number of answer scripts, the asynchronous queue-based inference mechanism is utilized for the orchestration of GPU-intensive OCR and LLM computations, with the help of the light-weight SQLite database storing all extracted text, results of evaluation and metadata.

A. Vision-Language OCR Module

The Vision-Language OCR module converts handwritten examination papers into readable text format, which can be analyzed semantically. The OCR module is not like regular OCRs since it uses the Florence-2 Vision-Language Model that has both visual and linguistic components to interpret handwritten text.

Given an input handwritten document,

$$I \in \mathbb{R}^{H \times W \times C}, \tag{1}$$

where H, W, and C denote the image height, width, and color channels, respectively. The first stage extracts meaningful visual features from the input image,

$$F = E(I), \tag{2}$$

where $E(\cdot)$ represents the Florence-2 visual encoder and F denotes the high-level visual feature representation. Rather than identifying individual characters, the encoder learns handwriting strokes, word structures, spacing patterns, and document layout, thereby generating a contextual representation of the handwritten document.

The extracted visual features are subsequently interpreted by the language decoder, which predicts the most probable text sequence,

$$\hat{T} = \arg\max_{T} P(T \mid F), \tag{3}$$

where $T = \{t_1, t_2, \ldots, t_N\}$ represents the generated text sequence. During decoding, each token is generated by considering both the visual features and the previously predicted tokens,

$$P(T|F) = \prod_{i=1}^{N} P(t_i \mid t_{<i}, F), \tag{4}$$

where $t_{<i}$ denotes the sequence of previously generated tokens. This autoregressive decoding strategy enables the model to preserve contextual consistency while recognizing handwritten text.

Consequently, the proposed module performs document-level understanding rather than isolated character recognition, making it robust to handwriting variations, noisy scans, and complex examination layouts. The extracted textual representation is subsequently forwarded to the constraint-aware LLM grading module for downstream semantic evaluation.

B. Constraint-Aware LLM Grading Module

The Constraint Aware LLM Grading Module uses semantic evaluation where the student's answer is matched with the reference answer provided by the teacher. Unlike traditional approaches of keyword matching for grading, the suggested approach involves the use of Llama 3 for reasoning on the conceptual semantics of the answer texts.

Let the extracted student response and reference answer be represented as

$$S = \{s_1, s_2, \ldots, s_n\}, \qquad R = \{r_1, r_2, \ldots, r_m\}, \tag{5}$$

where S and R denote the token sequences of the student and reference answers, respectively. Together with the grading prompt P, they are combined to form the contextual input

$$X = [P; R; S]. \tag{6}$$

The contextual input is first encoded into a semantic representation,

$$H = E(X), \tag{7}$$

where $E(\cdot)$ denotes the context encoding function of the Llama 3 model. The encoded representation is subsequently processed by the semantic decoder,

$$\phi = D(H), \tag{8}$$

where $D(\cdot)$ represents the semantic reasoning function and ϕ captures the conceptual similarity, contextual relevance, and completeness of the student response. Collectively, the semantic reasoning process is expressed as

$$\phi = f_{\mathrm{LLM}}(S, R, P) = D(E([P; R; S])) \,. \tag{9}$$

The final grading decision is obtained by

$$G = g(\phi, C), \tag{10}$$

where C denotes the examination constraints and $g(\cdot)$ represents the constraint-aware grading function. By jointly considering semantic understanding and evaluation constraints, the proposed module produces reliable and consistent grading results for handwritten descriptive answer scripts.

C. Asynchronous Inference and Data Management

Asynchronous Inference and Data Management is used to ensure efficient functioning of the proposed solution while ensuring that the evaluation data remains stored reliably. Due to the computational resource needs of both Florence-2 Vision-Language OCR model and Llama 3 semantic grading model, processing multiple answer scripts at the same time may cause the GPU resource conflicts. The solution to the problem is the use of asynchronous queuing scheduling approach by the proposed framework which will process answer scripts one by one.

Every uploaded answer script is queued for evaluation and graded independently using the OCR module and semantic grading module. This sequential scheduling avoids any conflict in GPU inference, optimizes the usage of resources, and allows the batch processing of handwritten answer scripts.

Once the evaluation is completed, the output text, evaluation outcomes, and metadata are stored in a SQLite database permanently. This database acts as a centralized storage medium for maintaining evaluation information. This helps in retrieving the results easily, auditing them and performing any future analysis on them. By combining asynchronous inference with persistent storage, we propose an efficient approach to automated evaluation of handwritten answers.

IV. Dataset and Implementation

A. Dataset Preparation

The effectiveness of the developed Vision-Language OCR module relies on the recognition of varied handwriting samples encountered in the handwriting of handwritten exam papers. As a way of improving the recognition of handwritten texts, IAM Handwriting Database has been chosen to be used as the main training database since it has been extensively used as the standard for HTR. In addition, multimodal document understanding has recently been introduced as an essential part of automatic assessment systems prompting the need to develop a Vision-Language OCR system [20].

IAM database consists of over 1,500 scanned handwritten pages by 657 different writers, including around 13,000 labeled text lines and above 115,000 word images. There is a high variability in handwriting styles, character spacing, strokes, and handwriting quality in this database which enables learning strong visual representations for practical handwritten documents.

Before the training process, the handwriting samples along with their ground truth transcriptions were prepared as image-text pair data suitable for training with the Florence-2 system. Validation, preprocessing, and formatting of the data ensured that the correlation between the handwritten data and the text was maintained. Such preparation helps the Vision-Language model learn the mapping between the handwriting data and its corresponding text, providing valid text data for automated answer evaluation using AI [16], [20].

B. Florence-2 Fine-Tuning

Though the Florence-2 model is pretrained on the general vision-language understanding task, its capability of recognizing handwritten texts from the unconstrained exam scripts is constrained. In order to increase the recognition accuracy of the model, it was fine-tuned through the IAM Handwriting Database that had been prepared before [18], [20].

The training dataset consists of image-text pairs (I, T), where I represents the handwritten image and T denotes its corresponding ground-truth transcription. During fine-tuning, the model learns the optimal parameters by minimizing the OCR sequence prediction loss,

$$\theta^* = \arg\min_{\theta} \mathcal{L}_{OCR}(\theta), \tag{11}$$

where

$$\mathcal{L}_{OCR} = -\sum_{i=1}^{N} \log P(T_i \mid I_i; \theta). \tag{12}$$

Here, $P(T_i \mid I_i; \theta)$ is defined as the likelihood of producing the right transcription from the handwritten input. The optimization criterion allows the model to improve the relationship between handwritten input and its corresponding text representation, which results in better recognition of different types of handwriting [13], [18].

The refined model is then incorporated into the Vision-Language OCR module proposed, making it possible to reliably extract hand-written text from scanned examination papers. The incorporation makes the model resistant to handwriting styles, uneven letter spacing, document scanning issues, among other problems associated with handwriting recognition, thus providing reliable input for semantic grading [16], [20].

TABLE II
OCR MODEL COMPARISON ON IAM HANDWRITING BENCHMARK

Model	CER (%)	Offline	Multimodal
Tesseract OCR	12.5	Yes	No
CRNN-CTC	5.45–8.1	Yes	No
TrOCR Base	2.89–3.5	Yes	No
Groq Vision API	∼3.0	No	Partial
Florence-2 (Proposed)	Competitive	Yes	Yes

C. System Implementation

This framework was implemented in a modular and offline-first design to guarantee a reliable and scalable evaluation of handwritten descriptive answers scripts. The framework incorporates three main modules: Vision-Language OCR, semantic grading with constraints, and asynchronous inference management [16], [20].

The backend was built with the use of Python and Flask, coordinating the entire process from the preprocessing of the documents to OCR detection, semantic analysis, and the management of the results using an execution pipeline. The model used for handwritten text extraction is called Florence-2. The local deployment of Llama 3, using Ollama, is responsible for semantic analysis without the need for any cloud-based services [19], [20].

The framework utilizes an asynchronous queue to schedule OCR and LLM inference tasks separately to ensure no GPU contention occurs during simultaneous operations. Results, text output, and metadata are permanently stored in an efficient and lightweight SQLite database for efficient retrieval, result tracking, and management. This makes the design flexible enough such that the different modules can work separately but in a coherent manner to accommodate future incorporation of other OCR and LLM models or grading techniques [16].

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. OCR Performance Evaluation

The suggested module of Vision-Language OCR was tested using the IAM Handwriting Database, a well-known dataset used in handwritten text recognition studies. As OCR quality influences the quality of semantic grading, recognition quality was examined in terms of handwriting style diversity and implementation features [20].

The optimized Florence-2 system was benchmarked against Tesseract OCR, CRNN-CTC, TrOCR Base, and Groq Vision API based on Character Error Rate (CER), off-line operation, and multimodal document understanding [16], [20].

The experimental results show that the suggested Florence-2 model exhibits comparable OCR accuracy, with the capacity to deploy in an offline setting and to process multimodal documents. This contrasts with traditional OCR systems that focus on text recognition or cloud services that require connectivity, and it offers a secure and private solution for evaluating handwritten answers [13], [18], [20]. The quantitative comparison of the evaluated OCR models is presented in Table II.

Table II demonstrates that Florence-2 is a multimodal document understanding system which, at the same time, provides high recognition accuracy and offline operation. Although Tesseract OCR, CRNN-CTC, and TrOCR Base offer offline inference but not multimodality, Groq Vision API offers multimodal reasoning but requires cloud connectivity. Thus, the proposed Florence-2 system can be used effectively for offline handwriting evaluation.

B. Semantic Grading Performance

To test the semantic grading capabilities of the proposed framework, experiments were carried out by comparing the OCR-extracted answers of students with the answers provided by the instructor with the help of Llama 3 grading module. Unlike traditional methods which work on the basis of keyword matching, language models conduct semantic evaluation by evaluating the concepts instead of matching lexical forms [5], [10], [18]. Thus, the proposed framework conducts conceptual evaluation via semantic reasoning.

Grading is conducted in an integrated way with respect to the grading question, reference answer, and the student's answer, focusing on semantic similarity as opposed to lexical similarity. This method increases robustness for paraphrased answers, synonyms, and differences in arrangement usually associated with descriptive tests [7], [9], [11].

As per the experimental outcomes, it is shown that the suggested semantic grading system is very much in accordance with the expectation of instructors with respect to consistency in grading the handwritten answers. Moreover, inclusion of fine-tuned Vision-Language OCR system helps avoid the errors due to OCR in the subsequent semantic evaluations [16], [19], [20].

C. Novelty

This solution solves all the shortcomings in the following ways: through the use of a refined version of Florence-2 Vision-Language OCR model along with a grading engine of Llama 3 with constraint awareness, asynchronous queue inference, and results management in SQLite database, providing an offline-first design.

VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed an AI-Based Automated Evaluation Framework for Handwritten Descriptive Answer Scripts by leveraging a fine-tuned Florence-2 Vision-Language OCR model and a constraint-aware Llama 3 semantic grading engine. Our framework can automate the whole evaluation process in terms of extracting handwritten text, conducting semantic evaluation, managing asynchronous inference, and storing results within an offline-first system. Experimental evaluation proved that our fine-tuned Florence-2 model attains comparable OCR accuracy while allowing for the semantic grading capability via context reasoning. Moreover, the proposed approach is scalable, efficient in GPU usage, and easily integrates OCR and LLM evaluation into educational assessment.

The future of our research will involve adapting the proposed framework to handle multilingual handwritten answer

TABLE III
PERFORMANCE COMPARISON OF OCR MODELS ON THE IAM HANDWRITING BENCHMARK

Model	CER	WER	Precision (%)	Recall (%)	F1-Score (%)	Avg. OCR Time (s/page)
Tesseract OCR	0.615	0.938	66.67	42.31	51.76	**0.268**
CRNN-CTC	0.519	1.438	61.11	74.04	66.96	2.805
TrOCR Base	0.192	0.563	83.62	93.27	88.18	8.221
Florence-2 (Proposed)	**0.019**	**0.250**	**100.00**	**98.08**	**99.03**	2.957

sheets, mathematics equations, tabular answers, and diagrams by means of advanced multimodal reasoning. Another possibility for scaling up our proposed framework is by using larger Vision-Language models and domain-specific Large Language models to increase grading accuracy in various academic subjects. Further research will also explore adaptive rubric learning, human-in-the-loop evaluations, and cross-institutional validation of such AI-driven assessments.

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